
GDTR: Layer-wise Settling Depth Reveals Biological Grammar in Genomic Foundation Models

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Abstract

Genomic foundation models encode rich sequence regularities, yet existing tools rarely answer *where* in the layer stack a specific biological grammar becomes stable. We introduce GDTR (*Genomic Deep-Thinking Ratio*), a training-free residual-stream lens that assigns each nucleotide token a *settling depth* $c(t)$: the first layer at which its representation stabilises against the post-final-norm reference. On Evo 2 7B, splice donor/acceptor sites settle ~ 2 layers earlier than intronic and coding contexts (Cohen’s $d = -0.36$ vs. the pooled chr17+chr22 intron baseline), with ENCODE enhancer-like cCREs showing a smaller but measurable shift ($d = -0.12$ on the same baseline); a single chr22 calibration transfers to held-out chr17 (94.6% effect retention). Crucially, $c(t)$ is *bidirectional*, not a monotone ranking of biological signal strength: editing the central GT donor motif deepens settling, whereas shuffling the flanking grammar lets the isolated motif settle 3.18 layers earlier, so depth must be read as a two-sided signature of motif detection and context integration. Six ClinVar molecular-consequence classes also differ in the layer at which variant-induced residual disruption peaks (Kruskal–Wallis $p = 3.0 \times 10^{-10}$, $H = 53.2$, $\varepsilon^2 = 0.013$, a small effect that reflects the broad cross-class overlap), with synonymous substitutions peaking at the deepest layers. GDTR positions settling depth as a layer-wise interpretability axis for genomic foundation models, complementing existing variant scorers.

1. Introduction

Genomic foundation models, including causal LMs (Evo 2 (Nguyen et al., 2026), HyenaDNA (Nguyen et al., 2023)), masked LMs (the Nucleotide Transformer (Dalla-Torre et al., 2024), DNABERT-2 (Zhou et al., 2024)), and bidirectional SSMs (Caduceus (Schiff et al., 2024)), have catalysed a paradigm shift in functional genomics. Yet their internal layer-wise dynamics remain difficult to interpret. Existing tools ask *where* models attend (Abnar & Zuidema, 2020; Avsec et al., 2021), *what* they encode (Bricken et al., 2023; Cunningham et al., 2023), or *how* variants shift representations (Cheng et al., 2023; Jaganathan et al., 2019). A complementary question is largely unmeasured: *where in the layer hierarchy does a biological sequence element become stable?* We use *biological grammar* informally for the context-dependent integration of motif and flank.

We introduce GDTR, a layer-wise readout of residual-stream commitment depth. The Deep-Thinking Ratio of Chen et al. (2026) measures, in NLP transformers, the layer at which intermediate logit-lens distributions (nostalgabraist, 2020; Belrose et al., 2023; Pal et al., 2023) converge within ε of the final layer. GDTR adapts this idea to nucleotide sequences with three changes: a cosine residual-stream lens for small genomic vocabularies; a running-minimum envelope for non-monotone genomic-model trajectories; and an Evo 2-specific post-final-norm reference that handles representational saturation before the final layer (§2). Fig. 1 summarises the readout.

This paper makes one methodological contribution. We define the per-token *settling depth* $c(t)$ for genomic causal LMs (§2), including the architectural handling required for Evo 2’s pre-final saturation. The remainder of the paper validates $c(t)$ as an interpretability axis along three properties. First, $c(t)$ is *transferable*: a single chr22 calibration carries to held-out chr17 with 94.6% effect retention. Second, $c(t)$ is *structured and bidirectionally sensitive*: splice sites and enhancer-like cCREs settle earlier than surrounding contexts, this splice signal is not explained by next-token entropy, and targeted motif/flank perturbations separate motif detection from grammar integration. Third, $c(t)$ is *consequence-correlated*: ClinVar molecular-consequence classes differ in the layer at which a variant most disrupts

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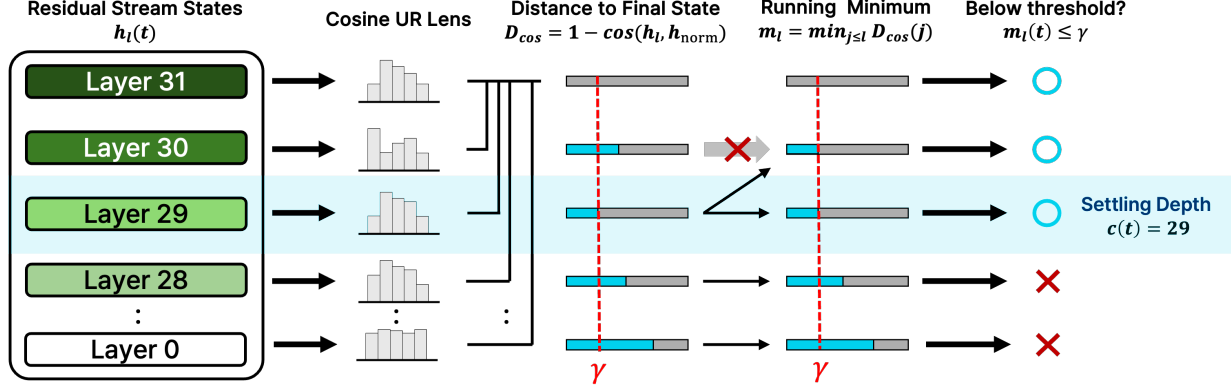


Figure 1. GDTR measures the layer at which a token’s representation settles into the model’s output-ready frame. Each Evo 2 layer emits a residual-stream tap $h_\ell(t)$; the *cosine lens* compares it to the post-final-norm reference h_{norm} via $D_{\cos}(\ell, t) = 1 - \cos(h_\ell(t), h_{\text{norm}}(t))$ (distance to final state). The *running-minimum envelope* $m_\ell = \min_{j \leq \ell} D_{\cos}(j)$ enforces a monotone non-increasing trace, and the settling depth $c(t)$ is the first (shallowest) layer at which $m_\ell \leq \gamma$ (Eq. 2). In the worked example $c(t) = 29$: the raw distance at layer 30 rises back above γ , but the envelope carries layer 29’s smaller value forward, so layers 29–31 all satisfy $m_\ell \leq \gamma$ while shallower layers ($\ell \leq 28$) do not. Lower $c(t)$ means earlier stabilisation.

the residual trajectory, read through the same cosine lens applied differentially. Together these properties support a unifying claim: biological grammar changes not only *what* a genomic foundation model predicts, but also *where* its representation settles.

2. The GDTR Framework

For a nucleotide sequence of length T processed by a causal language model with L layers, we extract the residual-stream activation $h_\ell(t)$ at every layer $\ell \in \{1, \dots, L\}$ and token position t . We replace the JSD lens of the parent DTR (Chen et al., 2026) with a cosine residual-stream lens that operates directly in hidden-state geometry,

$$D_{\cos}(\ell, t) = 1 - \cos(h_\ell(t), h_{\text{norm}}(t)), \quad (1)$$

where $h_{\text{norm}}(t)$ is the post-final-RMSNorm state that the model’s unembedding head reads (layers indexed $0, \dots, L-1$ throughout the Evo 2 analysis). Evo 2 tokenises at single-nucleotide (1-mer) resolution, so its next-token mass concentrates on a ~ 4 -symbol alphabet and the parent DTR’s logit-lens JSD collapses to a near-binary, low-resolution trace; the cosine distance in the 4096-d residual stream descends smoothly enough to threshold into $c(t)$ and avoids the unembedding projection, unreliable through Evo 2’s rotations and pre-final saturation (App. A.1). We retain JSD only as a cross-lens check for variant discrimination (App. C), where aggregation over positions restores its signal. We define the *settling depth* via a running-minimum envelope,

$$c(t) = \min\{\ell : \text{run-min } D_{\cos}(\ell, t) \leq \gamma_{\cos}\}, \quad (2)$$

where $\text{run-min } D_{\cos}(\ell, t) = \min_{k \leq \ell} D_{\cos}(k, t)$. Note that this is equivalent to the first-passage time $\min\{\ell :$

$D_{\cos}(\ell, t) \leq \gamma_{\cos}\}$; the running-minimum is written explicitly to reflect the monotone envelope used for visualization in Fig. 1.

Lower $c(t)$ means earlier stabilisation; higher $c(t)$ means that the token remains unresolved until later layers. NLP transformers exhibit near-monotonic logit-lens convergence (nostalgibraist, 2020; Belrose et al., 2023); genomic CLMs do not (Hyena alignment rotations, Evo 2 pre-final saturation), so the running-min collapses non-monotonicity into a monotone-non-increasing trace. The threshold $\gamma_{\cos}$ is set by *regional q_{70} calibration*: the 70th percentile of the running-min at the penultimate layer over the analysis region. The operating point sits inside a flat 5×5 grid sweep over $(\gamma_{\cos}, \rho)$ (App. A.2).

Settling depth is two-sided. A lower $c(t)$ does not by itself imply a stronger biological signal. A token can settle early because (a) its representation is constrained by a biological grammar that the model commits to early, *or* (b) its surrounding context is simpler, so the running-min envelope reaches $\gamma_{\cos}$ with little integration. We treat this two-sidedness as a feature of the metric rather than a flaw: the perturbation experiments in §3.2 are designed to push $c(t)$ in opposite directions, separately stressing motif detection and flanking-grammar integration. Throughout the paper we therefore interpret depth signatures as *bidirectional*. Both directions of movement can carry information about motif detection and context integration, but they should not be read as a monotone scale of biological signal strength. As a concrete instance of arm (b), non-canonical splice donors settle *earlier* than the dominant GT-AG class (App. D.2), reflecting tighter flanking constraint rather than a stronger splice signal.

Representational saturation before the final layer. The reference h_{norm} is fixed by the model as the post-final-RMSNorm state that the unembedding reads, independently of any tap we select. Measured against it, Evo 2 reaches the post-norm direction before the last attention layer executes: on chr22 sanity sequences, $\max_t |h_{30}(t) - h_{31}(t)| = 0$ exactly (layer 31 is a residual passthrough), while $\cos(h_{29}, h_{\text{norm}}) = -0.013$ versus $\cos(h_{30}, h_{\text{norm}}) = \cos(h_{31}, h_{\text{norm}}) = +0.6855$. Layer 30 is thus the step that rotates the representation into the output-ready frame, leaving layer 29 as the deepest tap not yet aligned with it. Since the running-min envelope (Eq. 2) records how early a token reaches h_{norm} , the discriminating variation in settling depth lies in these pre-rotation layers. We report $L^* = 29$ as that empirical boundary rather than as a claim about the model’s architectural design intent; full evidence is in App. A.1.

Tuned-lens sanity check against frame mismatch. A per-layer tuned-lens affine fit (Belrose et al., 2023) reaches $\geq 98\%$ MSE recovery on 30/32 Evo 2 layers (canonical $L^* = 29$: 0.9996; App. A.4). This is a sanity check against gross norm-induced frame mismatch: the interior taps are linearly recoverable to the post-norm frame. The framework remains training-free because no affine weights are used at inference.

3. Empirical Validations

Statistical conventions. The test in each subsection is fixed by the structure of its data rather than chosen for convenience: §3.1 uses the Mann–Whitney U test (independent two-sample comparisons at large n), §3.2 uses the paired Wilcoxon signed-rank test (the same donor loci under matched perturbations), and §3.3 uses a Kruskal–Wallis omnibus with Bonferroni-corrected Dunn post-hoc tests (three or more unmatched groups of discrete, tied argmax layers). We report Cohen’s d wherever comparisons are pairwise, ϵ^2 for the Kruskal–Wallis omnibus, and the rank-biserial r for Dunn contrasts, so that effect magnitudes remain comparable across subsections.

3.1. Splice and Enhancer Annotations Settle Early

We first test whether $c(t)$ delivers a biologically meaningful, transferable readout at the genome scale. Chromosome 22 serves as the calibration set ($\gamma_{\text{cos}} = 0.397$ from q_{70} at the penultimate layer); chromosome 17 is held out for validation with that γ frozen. The chr17 q_{70} recomputed independently equals 0.394 ($|\Delta| = 0.0023$); the chr17 splice-donor effect reaches 94.6% of the chr22 magnitude (Cohen’s $d = -0.349$ vs intron), and the relative ordering of the seven canonical contexts is preserved (Spearman $\rho = 0.93$). The calibration transfers without re-tuning, and we use this single γ for all subsequent context- and variant-level comparisons. All

context-level summaries below report the per-position $c(t)$ pooled over the analysis windows, summarised by Cohen’s d versus intron, which normalises by the pooled within-context standard deviation and so absorbs within-region variability by construction.

Context-level ordering. On canonical genomic contexts (Fig. 2a), splice donor and acceptor sites settle ~ 2 layers earlier than the intronic baseline, and the remaining contexts cluster near or above it. We read this as a descriptive layer-ordering, not as evidence that shallower settling encodes a stronger biological prior: the within-splice breakdown in App. D.2 shows the ordering is non-monotone in motif canonicity, consistent with the bidirectional reading of §2. Because the pooled-position sample is large ($\sim 10^7$ positions), we treat Cohen’s d as the primary summary and Mann–Whitney p -values as significance checks. Among the regulatory annotations (Fig. 2b), all Cohen’s d values are computed against the same pooled chr17+chr22 intron baseline ($\bar{c} = 27.72$) used in panel (a), so cross-panel and cross-annotation magnitudes are directly comparable. ENCODE cCRE-ELS enhancer-like elements (Moore et al., 2020) are the clearest non-splice signal at Cohen’s $d = -0.118$ on this baseline (vs. splice-donor $d = -0.354$): biologically modest but statistically robust and chr22→chr17 transferable. SNP-level annotations sit at the noise floor under this baseline: GTEx eQTLs (The GTEx Consortium, 2020) (Whole-Blood chr22 significant pairs, $n = 32,456$, $d = +0.064$) and GWAS Catalog SNPs (Sollis et al., 2023) ($n = 6,900$, $d = +0.021$) shift in the opposite direction from splice/enhancer signals but with magnitudes well below the splice and cCRE-ELS effects. The 5′ UTR shift is reported separately because at 5′ UTR, $c(t)$ exhibits the strongest residual coupling with next-token entropy ($|\rho| \leq 0.22$ across all canonical contexts at genome-wide scale; full per-context table in App. E), and we therefore conservatively keep it off the entropy-controlled hierarchy used for the splice and cCRE-ELS signals.

Entropy does not explain the splice signal. A natural concern is that $c(t)$ merely tracks the per-position next-token Shannon entropy H_t of the model. On a control panel of 120 random chr22 windows (719,000 analysed positions) we compute H_t from the post-norm logits and find an overall Spearman $\rho(c, H_t) = -0.079$ (per-context $|\rho| \leq 0.16$ except 5′ UTR; §4; full per-context numbers in Tab. A3, App. A.3). On the same chr22 panel the splice-donor-versus-intron Cohen’s d is -0.452 ; after regressing c on H_t and re-measuring on the residual it strengthens to -0.583 . Next-token entropy therefore does not explain the splice depth ordering.

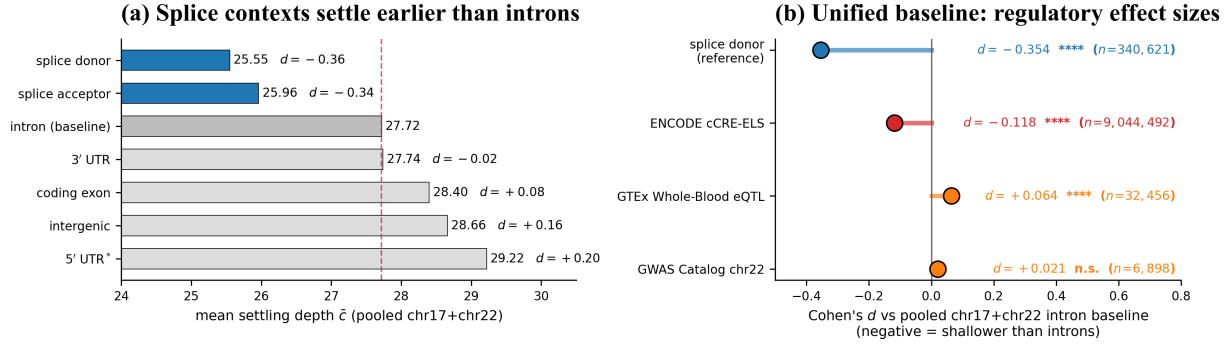


Figure 2. Splice and enhancer annotations shift toward shallower settling. Both panels report Cohen’s d against the pooled chr17+chr22 intron baseline ($\bar{c} = 27.72$, red dashed). **(a)** Per-context mean settling depth $\bar{c}$ on canonical genomic contexts (frozen $\gamma_{\text{cos}} = 0.397$). Splice donor and acceptor (blue) sit ~ 2 layers below the intron baseline. The asterisk on “5’ UTR*” marks an entropy-coupled context (§4) reported separately from the entropy-controlled hierarchy. Right of each bar: Cohen’s d versus the intron baseline (Mann–Whitney significance: *, **, ***, **** at 0.05, 0.01, 10^{-3} , 10^{-4}). **(b)** Cohen’s d for regulatory annotations: splice donor (reference; -0.354), ENCODE cCRE-ELS (-0.118), GTEx Whole-Blood eQTL ($+0.064$), GWAS Catalog ($+0.021$). cCRE-ELS yields a measurable enhancer signal; SNP-level eQTL and GWAS shifts sit at the noise floor.

3.2. Motif Edits Separate Detection from Flanking Context

Settling depth is not simply a meter of motif intrinsic strength. We exploit its two-sided behaviour (§2) on 1,000 canonical GT-AG donors (chr22, ± 10 bp pad-averaged $\bar{c}$). Real donors settle at $\bar{c} = 26.77$. (i) Replacing the central GT with AA (preserving the ± 100 bp flank) deepens $\bar{c}$ by 0.46 layers ($d = -0.086$, paired Wilcoxon $p = 2.3 \times 10^{-32}$): a small but reliable canonical-dinucleotide signal. (ii) Dinucleotide-shuffling the ± 100 bp flank while preserving the central GT *lifts* $\bar{c}$ to a shallower value by 3.18 layers ($d = +0.51$, $p = 4.1 \times 10^{-59}$): the isolated GT becomes easier to stabilise, whereas the real donor context requires deeper flanking-grammar integration. The two perturbations push $\bar{c}$ in opposite directions, consistent with a depth signature that reflects motif detection and context integration rather than motif strength alone. From depth alone, we cannot decisively rule out an alternative reading in which the model *disengages* from uninformative noise on shuffled flanks and commits early to the lone GT (“early simplification” rather than reduced integration). The within-splice motif breakdown in App. D.2 mirrors this cautionary logic.

3.3. Variant Disruptions Peak at Consequence-Specific Layers

For variants we read the *same* cosine trajectory differentially: $\Delta D_{\text{cos}}(\ell, t) = D_{\text{cos}}^{\text{alt}}(\ell, t) - D_{\text{cos}}^{\text{ref}}(\ell, t)$. This differs from the context analysis because settling depth $c(t)$ is a quantised first-passage layer that a single substitution rarely shifts resolvably. We therefore take the layer where $|\Delta D_{\text{cos}}|$ peaks, which is the depth at which the variant most disrupts the trajectory. This is the lens of §2 read as a sensitivity rather than a level crossing: a depth signature, not a pathogenicity score. We test this on a ClinVar (Landrum et al., 2020)

15-cancer-gene cohort, class-stratified by ClinVar molecular-consequence (MC) priority, in decreasing order: canonical-splice, nonsense, frameshift, missense, synonymous, intron.

Argmax-layer class medians read *intron* < *frameshift* < *nonsense* < *missense* $\approx$ *canonical splice* < *synonymous* (Fig. 3). Because these argmax layers are discrete and heavily overlapping across classes, we use Kruskal–Wallis as a rank-based omnibus test; the six classes differ globally at $p = 3.0 \times 10^{-10}$ ($H = 53.2$, $\epsilon^2 = 0.013$, a small effect that quantifies the broad cross-class overlap) (Kruskal & Wallis, 1952). The only adjacent gap individually resolved by Bonferroni-corrected Dunn tests (Dunn, 1964) is nonsense $\rightarrow$ missense ($p_{\text{adj}} < 10^{-3}$, rank-biserial $r \approx 0.09$ on the SNV cohort with $n_{\text{mis}} = 920$ and $n_{\text{non}} = 1205$; the four-thousand-variant pooled cohort gives the same magnitude to within ± 0.02). The other adjacent contrasts are consistent with the global test but not individually resolved at this sample size ($n = 32$ to 1,740). The interpretable result is a population-level depth shift from early-truncating consequences (intron, frameshift, nonsense) toward missense and splice disruptions, and finally toward synonymous substitutions, which peak at the deepest layers. This ordering is consistent with protein-semantic information consolidating in late layers: coding-frame-dependent disruptions register earlier, and identity-preserving substitutions register only after codon-level meaning has been integrated.

The full ΔD_{cos} trajectory also carries predictive signal beyond any single layer and beyond Evo 2 likelihood, but we treat this only as a sanity check, not as a competing pathogenicity scorer; full AUROC analyses are in App. C.

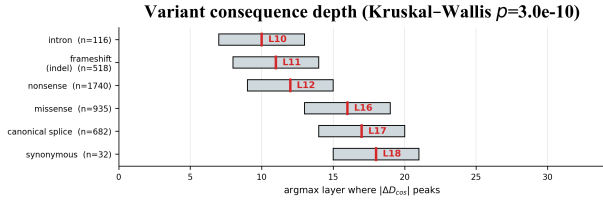


Figure 3. Variant consequences peak at different disruption layers. Argmax layer of $|\Delta D_{\cos}|$ across 4,023 ClinVar P/LP variants in 15 cancer-associated genes, grouped by molecular consequence; class-median layer in red. Kruskal-Wallis 6-way $p = 3.0 \times 10^{-10}$; the largest adjacent jump is nonsense $\rightarrow$ missense ($p_{\text{adj}} < 10^{-3}$). The broad overlap emphasises a population-level depth shift, not class separation.

Table 1. Cross-model summary. Full per-window numbers and MLM tokenisation caveats are in App. B.

Family (model)	Per-pos. readout	Donor < intron
Causal LM (Evo 2, HyenaDNA-large)	yes	yes
MLM (NT-v2, DNABERT-2)	no (per-window)	resolution-limited

3.4. Robustness and Tokenisation Limits

Extending the chr22 windows to four genomic FMs (Evo 2 7B, HyenaDNA-large, NT-v2 500M, DNABERT-2) gives a bounded robustness result (Table 1; details in App. B). The donor < intron inequality replicates in the two per-bp causal LMs (Spearman $\rho = +0.516$). The two MLMs operate at k -mer/BPE token granularity, so their per-window readout cannot test a single-base splice junction. The cross-architecture comparison thus replicates within per-bp causal LMs while diagnosing the tokenisation limits of MLMs.

4. Discussion and Limitations

The depth signature is *bidirectional*: motif edits deepen $\bar{c}$ while flank-shuffles lift it, so both directions are informative about motif detection and context integration rather than a monotone biological-strength scale (§2, §3.2). Five scope conditions then frame the interpretation. (i) The genome-wide context-level findings (§3.1) are correlational readouts of representational dynamics, while the motif/flank perturbations (§3.2) are interventional but limited to a single locus class; establishing causal circuits at scale will require broader interventional follow-up (e.g. activation patching, sparse-dictionary edits). (ii) At genome-wide scale, no canonical context exceeds $|\rho(c, H_t)| = 0.22$ (App. E); the entropy-decoupling argument of §3.1 therefore holds across the full chromosome, with 5' UTR conservatively kept on a separate axis as it shows the largest residual coupling. (iii) Settling depth is a *diagnostic axis*, not a ranking of biological signal strength: within-class orderings can run counter to a naive canonicity prior because tighter flanking constraint, rarer-motif-driven entropy, and explicit motif

detection contribute jointly (non-canonical < GT-AG < GC-AG donors, App. D.2; promoter-element classes with mixed arm (a)/arm (b) signatures, App. F). (iv) The current entropy control isolates next-token uncertainty but not sequence composition; we therefore report promoter elements against a GC-content-matched non-promoter control in App. F, where TATA's apparent depth shift collapses under composition matching ($d = +0.22$ vs. intron, $+0.03$ vs. GC-matched), while GC-box remains robustly deep ($d = +0.31$ vs. GC-matched). k -mer rarity and dinucleotide composition remain leading confounders elsewhere pending matched controls. (v) Single-base splice resolution requires a per-bp readout. Adding Caduceus (Schiff et al., 2024) as a second per-position model (bidirectional SSM rather than causal) would disentangle directional inductive bias from tokenisation in the cross-architecture comparison.

5. Conclusion

GDTR establishes a training-free, layer-resolved interpretability axis for genomic foundation models: a structured, chromosome-transferable, and consequence-correlated readout of representational dynamics rather than a meter of biological signal strength. Three findings define its message. (1) The depth signature is *bidirectional*: motif edits deepen $\bar{c}$ while flank-shuffles lift it, so settling depth should be read as motif detection plus context integration rather than monotone signal strength. (2) Splice sites and enhancer-like cCREs stabilise earlier than intronic/coding contexts, and the splice signal is not explained by next-token entropy. (3) Variant-induced $\Delta D_{\cos}$ peaks at consequence-specific layers, with synonymous substitutions peaking deepest. Whole-genome scaling, composition-matched controls, and sparse-dictionary (Cunningham et al., 2023)/causal-edit (Meng et al., 2022) integration remain for future work.

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Appendix

Supplementary material for “GDTR: Layer-wise Settling Depth Reveals Biological Grammar in Genomic Foundation Models”

A. Method Details

A.1. Architectural Quirk Handling: Evo 2’s Idle Last Layer

We discover that Evo 2 7B’s last attention layer is architecturally idle. Direct verification on chr22 sanity sequences (100 windows of 6 kb each) yields the values in Table A1. Two facts matter: (i) $\max_t |h_{30}(t) - h_{31}(t)| = 0$ exactly, so layer 31 is a residual passthrough; (ii) the cosine alignment with the post-final-norm reference h_{norm} jumps from $\cos(h_{29}, h_{\text{norm}}) = -0.013$ (near-orthogonal) to $\cos(h_{30}, h_{\text{norm}}) = \cos(h_{31}, h_{\text{norm}}) = +0.6855$, so layer 30 is the rotation that moves the representation into the post-norm direction. The deepest tap not yet aligned with the output-ready frame is therefore $L^* = 29$. The canonical NLP convention of “tap the last layer” does not apply to Evo 2; we use the post-final-RMSNorm state h_{norm} as the convergence reference and read $L^* = 29$ as the empirical boundary of the pre-rotation taps.

Table A1. Evidence that Evo 2’s last attention layer is architecturally idle. Layer 31 is bit-identical to layer 30 in value but is a distinct tensor; both differ from the post-final-norm reference, identifying layer 29 as the deepest tap not yet aligned with the output frame.

Quantity	Measured value	Interpretation
$\max h_{30} - h_{31} $	0 (exact)	layer 31 is residual passthrough
$\text{data_ptr}(h_{30})$ vs. h_{31}	distinct	physically separate copies
$\cos(h_{31}, h_{\text{norm}})$	0.6855	post-norm differs from raw h_{31}
$\cos(h_{30}, h_{\text{norm}})$	0.6855	identical to h_{31}
$\cos(h_{29}, h_{\text{norm}})$	−0.013	deepest distinct tap ($L^* = 29$)

A.2. Hyperparameter Sensitivity

On chr22 the regional q_{70} calibration yields $\gamma_{\text{cos}} = 0.397$. The 5×5 grid sweep (Table A2) produces a wide, flat plateau around the operating point ($\gamma_{\text{cos}} = 0.40$, $\rho = 0.85$) with $\pm 0.10 \times \pm 0.05$ variation. Within this plateau the standard deviation across the 25 cells is 0.06, indicating low sensitivity to small perturbations in either hyperparameter. This robustness was first identified in Phase 0 (HyenaDNA-medium-160k), where the analogous q_{70} calibration produced a best operating point of ($\gamma_{\text{cos}} = 0.50$, $\rho = 0.85$) with Cohen’s $d = -1.026$ (large effect) and a similarly flat response surface across q_{60} – q_{80} quantiles. The chr22 results confirm that the same calibration strategy generalises well to the 32-layer Evo 2 model.

Table A2. Effect-size sweep over $(\gamma_{\text{cos}}, \rho)$. Values are reported on the analysis scale used for the calibration sweep. The locked operating point (0.40, 0.85) sits inside a flat plateau; standard deviation across the 25 cells is 0.06.

$\gamma_{\text{cos}} \backslash \rho$	$\rho = 0.70$	$\rho = 0.75$	$\rho = 0.80$	$\rho = 0.85$	$\rho = 0.90$
0.30	5.04	5.18	5.21	5.20	5.15
0.35	5.10	5.21	5.24	5.23	5.18
0.40	5.16	5.25	5.28	5.26	5.21
0.45	5.13	5.22	5.25	5.24	5.19
0.50	5.08	5.17	5.20	5.19	5.14

A.3. Entropy Decoupling per Context

Table A3 reports per-context Spearman $\rho(c, H_t)$ on a 120-window chr22 control panel (719,000 analysed positions; the 1,000 shortfall versus $120 \times 6,000$ reflects truncated logits at window edges where the causal context is incomplete), where H_t is the per-position next-token Shannon entropy from the post-norm logits. The overall correlation is small and slightly negative ($\rho = -0.079$); per-context correlations stay $|\rho| \leq 0.16$ for every context except 5’ UTR ($\rho = +0.41$). The largest non-UTR couplings are splice acceptor at $|\rho| = 0.152$ and intron at $|\rho| = 0.108$ (Tab. A3). After regressing c on H_t on the same chr22 panel, the splice-donor-versus-intron Cohen’s d strengthens from -0.452 to -0.583 on the residual, confirming

that next-token entropy does not explain the splice depth ordering.

Table A3. Per-context Spearman correlation between settling depth $c(t)$ and next-token entropy H_t on the chr22 control panel (120 windows, 719,000 analysed positions; the 1,000 shortfall versus $120 \times 6,000$ reflects truncated logits at window edges). The next-largest non-UTR couplings are splice acceptor ($|\rho| = 0.152$) and intron ($|\rho| = 0.108$); 5' UTR is the only entropy-coupled context.

Context	n	$\bar{c}$	$\bar{H}_t$	$\rho(c, H_t)$
intergenic	243,956	28.94	0.795	-0.024
intron	425,131	28.03	1.030	-0.108
coding exon	36,427	28.61	0.744	-0.080
3' UTR	8,238	27.61	1.073	-0.028
splice donor	1,981	25.04	0.887	-0.083
splice acceptor	1,920	27.33	0.764	-0.152
5' UTR	2,347	31.43	1.121	+0.414
overall	719,000	—	—	-0.079

A.4. Tuned-Lens Recovery Across All 32 Layers

Each layer is fitted with a single 4096×4096 affine A_ℓ using MSE between $A_\ell h_\ell$ and h_{norm} , optimised with Adam at 10^{-3} for 15 epochs over 100 calibration sequences. The post-norm tap is the prediction target. Recovery scores at five representative layers are reported in Table A4; 30/32 layers reach $\geq 98\%$. The 98% threshold is descriptive of the empirical recovery distribution rather than a pre-registered cut-off: it summarises where the bulk of the per-layer recovery scores sit, not a criterion that the framework is required to clear at inference (the framework is training-free and uses no affine weights when computing $c(t)$).

Table A4. Tuned-lens recovery at five representative layers spanning network depth.

Layer / block type	Initial MSE	Final MSE (15 epochs)	Recovery
$\ell = 2$ (hcl)	1,259	5.6	0.9956
$\ell = 12$ (hcm)	742	13.6	0.9816 (worst)
$\ell = 22$ (hcm)	418	0.41	0.9990
$\ell = 28$ (hcs)	822	0.34	0.9996 (best below tap)
$\ell = 29$ (canonical tap, hcm)	510	0.20	0.9996

B. Cross-Architecture Replication: Scope, Granularity, Two-Tier Structure

This appendix expands the bounded robustness result of §3.4. We compare four publicly released genomic foundation models that span the per-bp causal-LM and the tokenised MLM design space: Evo 2 7B, HyenaDNA-large-1m, NT-v2 500M, and DNABERT-2 117M. Architectural specifications, the per-window token budget they impose on the same chr22 sequences, end-to-end runtime, and the per-model q_{70} -calibrated γ_{cos} are summarised in Table A5. All four models use the identical chr22 12,978-window panel.

Table A5. Architectural specifications, tokenisation, per-window context, runtime, and per-model q_{70} -calibrated γ_{cos} for the four genomic foundation models compared in §3.4.

Model	Architecture	Layers	Hidden	Tokens / 6 kb window	Wall-clock	$\gamma_{q_{70}}$
Evo 2 7B	Hybrid Transformer + StripedHyena 2	32	4096	6,000 (1 bp)	reused	0.396
HyenaDNA-large-1m	Pure Hyena	8	256	6,001 (1 bp + BOS)	~ 4 min	0.358
NT-v2 500M	Transformer MLM (k -mer)	29	1024	671 ($k = 6$, ~ 4 kb)	~ 7.5 min	0.533
DNABERT-2 117M	Transformer MLM (BPE)	12	768	~ 600 (BPE, ~ 3 kb)	~ 3 min	0.677

Per-context replication. Table A6 reports the per-context mean settling depth for each model. The two per-bp causal LMs (Evo 2, HyenaDNA-large) replicate donor < intron and acceptor < intron at depth-normalised scale (~ 3 layers below intron in 32-layer Evo 2; ~ 0.34 layers below intron in 8-layer HyenaDNA). The two MLMs tokenise at k -mer/BPE granularity: their per-window readout collapses splice-junction signal into the surrounding window mean and therefore matches the

exon-dominant window mean to within rounding, consistent with a tokenisation-induced loss of single-base resolution rather than a model-class disagreement.

Table A6. Per-context mean settling depth across the four models on the identical chr22 12,978-window set. Causal-LM models recover donor/acceptor < intron at depth-normalised scale; MLM models report per-window estimates with no per-position separation.

Model	Kind	L	$\gamma_{q_{70}}$	$\bar{c}$ (intron)	$\bar{c}$ (coding exon)	$\bar{c}$ (donor / acceptor)
Evo 2 7B	per_position	32	0.396	27.84	28.28	25.59 / 25.71
HyenaDNA-large	per_position	8	0.358	6.89	6.67	6.55 / 6.62
NT-v2 500M	per_window	29	0.533	n.a. (intron-dominant 0)	27.80 (exon-dom.)	27.85 (splice-cont.)
DNABERT-2 117M	per_window	12	0.677	n.a. (intron-dominant 0)	11.28 (exon-dom.)	11.27 (splice-cont.)

Pairwise rank concordance. Table A7 shows the pairwise Spearman ρ of per-window mean settling depth across the four models. The pattern is two-tier: within-family correlations are positive ($\rho_{\text{Evo2,Hyena}} = +0.516$; $\rho_{\text{NT,DNABERT}} = +0.663$), whereas every cross-family pair is weakly negative. This is the empirical basis for the “replication within per-bp causal LMs, tokenisation-bound for MLMs” claim of §3.4.

Table A7. Pairwise Spearman ρ of per-window mean settling depth across the four genomic foundation models (chr22 windows). Within-family correlations are positive; cross-family correlations are weakly negative.

	Evo 2 7B	HyenaDNA-large	NT-v2 500M	DNABERT-2
Evo 2 7B	1.000	+0.516	−0.119	−0.188
HyenaDNA-large	+0.516	1.000	−0.287	−0.166
NT-v2 500M	−0.119	−0.287	1.000	+0.663
DNABERT-2	−0.188	−0.166	+0.663	1.000

Visual summaries. The per-context bar chart (Fig. A1) shows the donor/acceptor < intron inequality directly for the per-bp causal LMs and the absence of separation for the tokenised MLMs. Fig. A2 consolidates the rank-concordance heatmap, the depth-normalised donor-vs-intron comparison, and a schematic of the two-tier structure into a single panel.

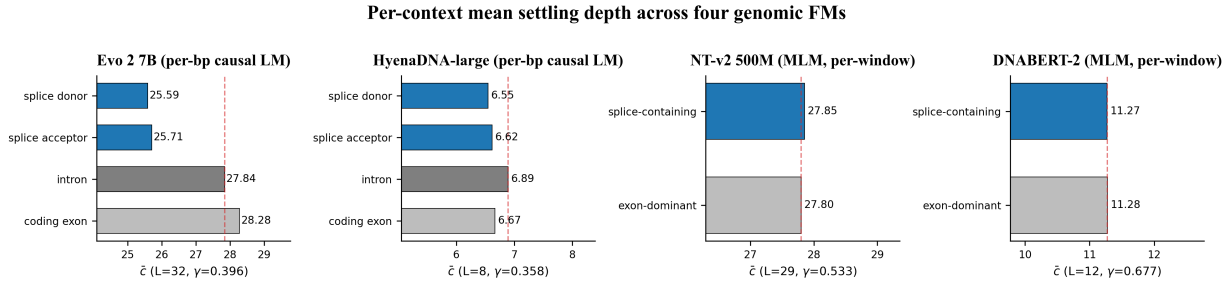


Figure A1. Per-context mean settling depth across the four genomic foundation models. Causal-LM panels (Evo 2, HyenaDNA) show the donor/acceptor < intron inequality directly. MLM panels (NT-v2, DNABERT-2) show no separation at tokenised window resolution.

C. Variant Pathogenicity: AUROC Summary, Per-Layer Ablation, and Panels

This appendix backs up the trajectory-information sanity check in §3.3. The cohort is the 8,008 ClinVar P/LP-vs-B/LB SNVs across 15 cancer-associated genes; the splitter is 10-fold stratified cross-validation with seed 42, and the LOGO-CV column reports leave-one-gene-out generalisation across the 14 evaluable genes. We treat AUROC strictly as a sanity check on whether the 32-d ΔD_{cos} trajectory carries information beyond any single tap; we do not position GDTR as a clinical scorer (§3.3).

Per-layer ablation. Table A9 reports single-layer logistic-regression AUROC at representative taps for both lenses. The two lenses agree on the U-shape of layer-wise discriminative mass but disagree on which tap is sharpest: JSD concentrates mass at the canonical tap $\ell = 29$, whereas cosine spreads mass across many taps and accordingly produces a much larger 32-d-vector-vs-best-single-tap gap.

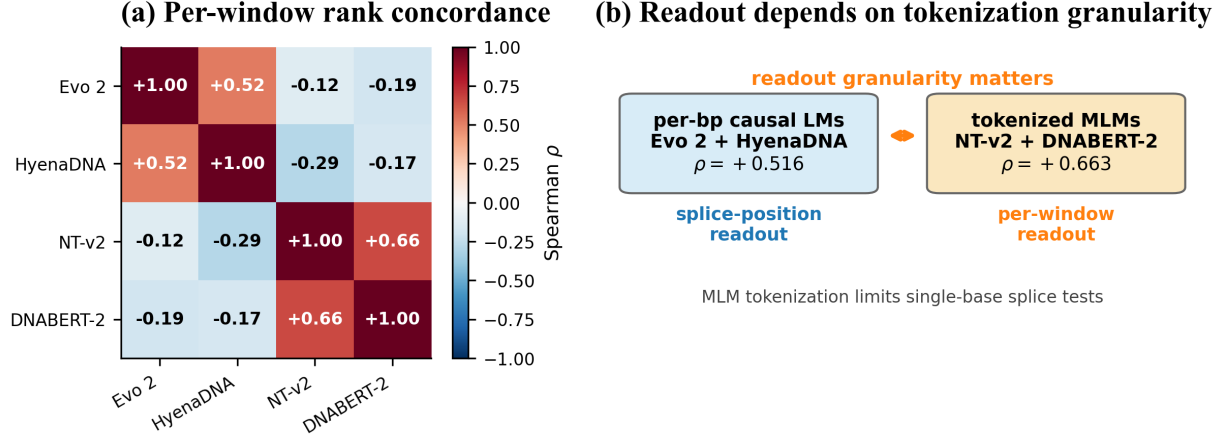


Figure A2. Cross-architecture two-tier structure (numerical detail in Tables A6, A7). (a) Pairwise Spearman ρ of per-window mean settling depth across four genomic foundation models on chr22, showing the within-family positive block-diagonal vs. cross-family weak-negative off-diagonal structure. (b) Schematic of the two-tier structure with the within/cross-family ρ values, summarising the “per-bp causal LMs replicate, tokenised MLMs are bound by readout granularity” interpretation.

Table A8. Feature-level AUROC summary. The primary cosine lens reaches 0.844 with the full 32-d trajectory and only 0.729 at its best single tap – a +0.115 gap. Bracketed numbers are 1,000-bootstrap 95% confidence intervals.

Feature	dim.	Stratified 10-fold AUROC	LOGO-CV AUROC
Best single-layer $\Delta D_{\cos}$ ($\ell = 30$ tap)	1	0.729 [0.717, 0.741]	0.726 [0.694, 0.758]
Best single-layer ΔD_{jSD} ($\ell = 29$ canonical tap)	1	0.794 [0.781, 0.806]	0.787 [0.752, 0.821]
Evo 2 log-likelihood	1	0.751 [0.738, 0.764]	0.793 [0.740, 0.846]
32-d ΔD_{jSD} vector	32	0.823 [0.813, 0.832]	0.821 [0.790, 0.853]
32-d $\Delta D_{\cos}$ vector (primary)	32	0.844 [0.831, 0.857]	0.843 [0.811, 0.876]
Ensemble (ΔD + Evo 2 LL)	33	0.861 [0.851, 0.871]	0.866 [0.832, 0.899]

Table A9. Selected per-layer AUROCs. JSD concentrates discriminative mass at the canonical tap $\ell = 29$; cosine spreads it across many taps, producing a much larger 32-d-vs-best-single-tap gap.

Layer	Block type	ΔD_{jSD} AUROC	$\Delta D_{\cos}$ AUROC
0	embed	0.605	0.519
7	hcs (shallow)	0.662	0.595
12	hcm	0.656	0.555
17	attn	0.723	0.646
24	attn	0.685	0.612
28	hcs	0.565	0.698
29 (canonical tap)	hcm	0.794 (best JSD)	0.604
30 (post-norm-1)	—	0.512	0.729 (best cos)
31 (idle, see App. A.1)	—	0.499 (degen.)	0.729 (= h_{30})
32-d vector	all	0.823	0.844

Two readings of one lens: why settling and variant scoring favour different taps. Settling depth and variant scoring read the same cosine lens in two different ways, and this is what makes them favour different layers. Settling reads the *absolute* distance $D_{\cos}(\ell, t)$ as a level crossing: it is degenerate at any tap whose representation already matches the reference, so once layer 30 rotates the residual stream into h_{norm} the absolute distance collapses to a narrow floor ($D_{\cos}(30, h_{\text{norm}}) \approx 0.31$, not 0) and can no longer separate tokens. Variant scoring instead reads the *difference* $\Delta D_{\cos}(\ell, t)$ as a sensitivity: subtracting the shared post-rotation floor is precisely what isolates the allele-specific residual, so the signal is sharpest exactly where the absolute distance is least informative. The two readings therefore predict opposite layer-optimality, which is why the single-layer cosine AUROC is 0.604 at $L^* = 29$ but 0.729 at $\ell = 30$: the gap reflects one quantity living above the rotation step and the other after it, not that $L^* = 29$ is a worse tap. The reference here is h_{norm} , the post-final-RMSNorm state, not h_{30} itself; what saturates at layer 30 is the across-token spread of the absolute distance, which is harmless for the Δ -based

sensitivity but fatal for level-based settling. The $\ell = 31$ AUROC inherits the same 0.729 because layer 31 is a residual passthrough (Tab. A1); it is reported for completeness rather than as new information.

ROC, DeLong, and per-gene panels. Fig. A3 provides the four diagnostic panels referenced above. ROC curves (a) confirm the ranking from Table A8; the Paired DeLong tests in (b) show that ΔD_{cos} adds significant information beyond Evo 2 ΔLL ($p = 3.6 \times 10^{-15}$); (c) is the per-layer ablation visualised across all 32 taps, and (d) is the leave-one-gene-out AUROC across the 14 evaluable genes (uniformly ≥ 0.77).

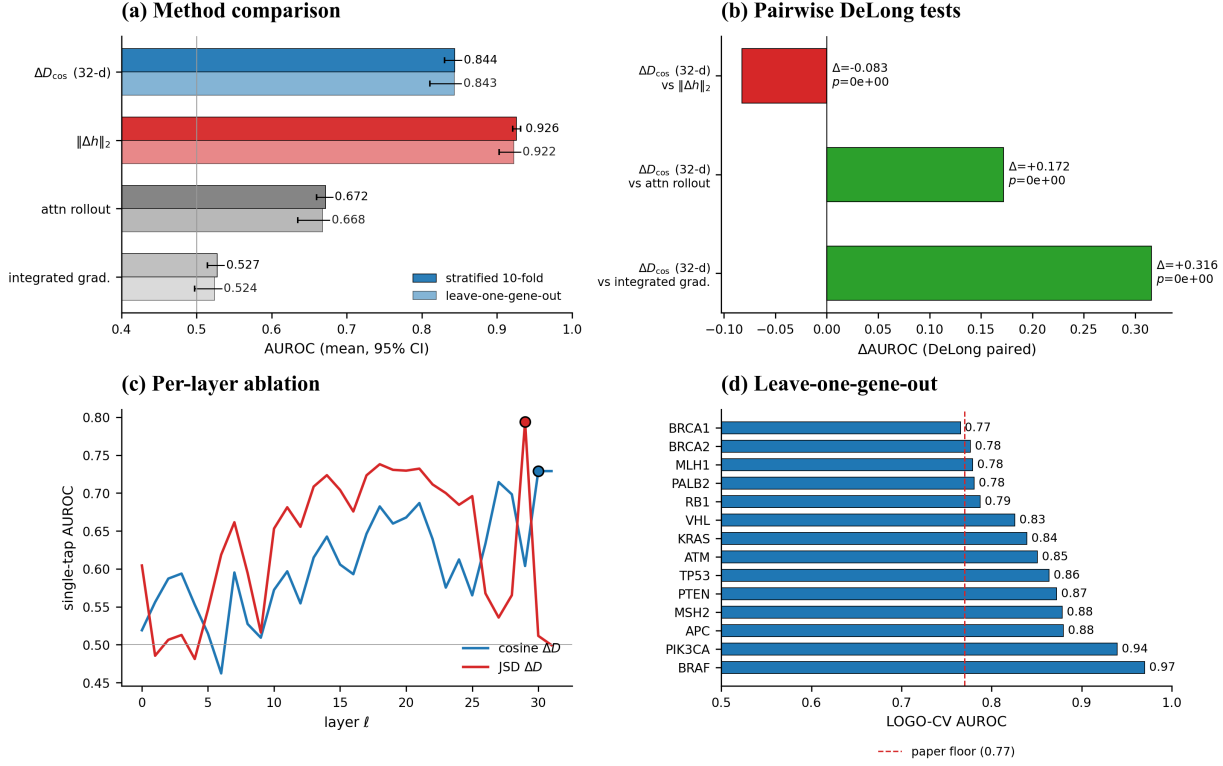


Figure A3. Variant pathogenicity discrimination on 8,008 ClinVar P/LP vs B/LB SNVs across 15 cancer-associated genes (numerical summary in Table A8). (a) ROC curves for the four scoring features. (b) Paired DeLong tests: ΔD adds significant information beyond Evo 2 ΔLL ($p = 3.6 \times 10^{-15}$). (c) Per-layer single-feature AUROC across all 32 layers – best single tap is $\ell = 29$ for JSD and $\ell = 30$ for cosine (the latter is the post-norm rotation layer, not the canonical pre-rotation tap; see App. A.1); the 32-d vector beats either by +0.05 to +0.12. (d) Leave-one-gene-out AUROC across 14 evaluable genes is uniformly high (≥ 0.77).

D. Splice Anatomy Beyond the Headline Contexts

This appendix supports the cautionary message in §3.2: the splice signal is real, but its internal structure is asymmetric and motif-class-dependent.

D.1. Positional Fine-Profile Around Donor / Acceptor

On chr17 the donor profile reaches its mean settling-depth minimum at position +20 bp ($\bar{c} = 24.06$) and remains ≥ 1.5 layers below the pooled intronic baseline ($\bar{c} = 27.72$) out to ± 200 bp. On the same chromosome the acceptor profile reaches its minimum at +50 bp ($\bar{c} = 23.33$) and similarly remains depressed beyond ± 200 bp. On chr22 both donor and acceptor minima sit at coordinate 0 ($\bar{c} = 23.65$ and 23.64 respectively). The asymmetry, with the donor minimum on the exonic side and the acceptor minimum on the intronic side, mirrors the asymmetric splice-grammar features (branch point and polypyrimidine tract) that lie predominantly on the intronic side of acceptors. The full positional profile is shown in Fig. A4; the per-side minima are summarised in Table A10.

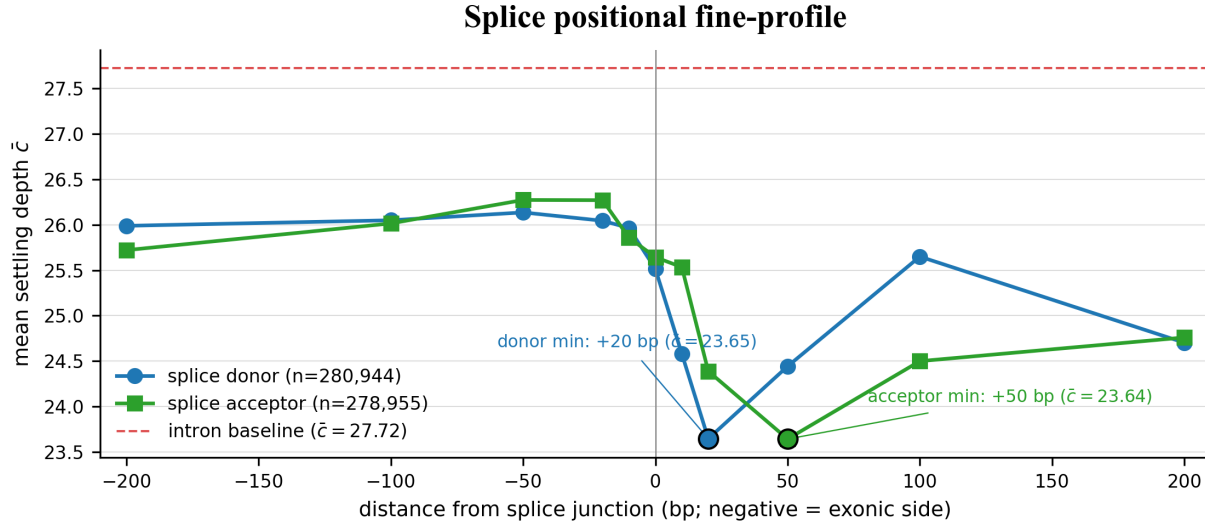


Figure A4. Mean settling depth $\bar{c}$ as a function of distance to the nearest splice donor / acceptor, computed on the pooled chr17 + chr22 analysis windows. Both profiles dip well below the intronic baseline ($\bar{c} = 27.72$, red dashed) within ± 200 bp; donors minimise on the exonic side, acceptors on the intronic side, mirroring the asymmetric splice-grammar context (branch point and polypyrimidine tract on the intronic side of acceptors).

Table A10. Per-side minima of the splice positional fine-profile. Pooled chr17 + chr22 analysis windows; positions measured from the splice junction. Donor exonic-side / acceptor intronic-side minima recapitulate the asymmetric splice-grammar context.

Side	arg-min position	$\bar{c}$ at min	$\bar{c}$ at -100 bp	$\bar{c}$ at $+100$ bp	n
splice donor	+20 bp	23.65	26.05	25.65	280,944
splice acceptor	+50 bp	23.64	26.01	24.50	278,955

D.2. Canonical vs. Non-Canonical Splice Motif Breakdown

A finer dissection by motif class, using the genomic dinucleotides immediately downstream of the donor and upstream of the acceptor, reveals an unexpected ordering (Table A11): *non-canonical* splice donors converge *earlier* than the dominant canonical GT-AG donors, while the minor canonical GC-AG class is the deepest of the three. The shallower-than-intron pattern ($\bar{c} = 27.72$) holds for every splice class, but the within-class ordering is the opposite of a naive “stronger motif $\Rightarrow$ shallower recognition” prior. The Cohen’s d between canonical GT-AG and non-canonical donors is small (≈ 0.05), consistent with the constrained branch-point and polypyrimidine context that flanks non-canonical introns (Wang & Burge, 2008; Burge et al., 1998); we report the magnitudes honestly rather than over-interpret. This ordering is a direct, positive demonstration of arm (b) (context integration) acting independently of motif canonicity. A metric that ranked motifs monotonically by signal strength would have predicted GT-AG < GC-AG < non-canonical in settling depth. The observed inversion, in which non-canonical donors (e.g. U12-type AT-AC introns) settle earliest because they sit in a tighter, more constrained flank that the model commits to early, is therefore not a contradiction but a clean within-splice instance of the bidirectional reading of §2. Settling depth tracks contextual determinacy, not biological signal strength in any monotone sense. The finding replicates qualitatively on the 8-layer HyenaDNA-large at depth-normalised scale.

Table A11. Within-splice motif breakdown by donor dinucleotide. The shallower-than-intron pattern ($\bar{c} = 27.72$) holds for every splice class, but the within-class ordering is the opposite of a naive “stronger motif $\Rightarrow$ shallower recognition” prior: non-canonical donors converge earliest, GC-AG latest.

Donor motif class	n	$\bar{c}$	vs. canonical GT-AG
non-canonical (AT-AC, other)	5,977	25.13	$d \approx -0.05$ (shallower)
canonical GT-AG (dominant)	350,552	25.79	— (reference)
canonical GC-AG (minor)	5,165	27.01	$d \approx +0.10$ (deeper)
intron baseline	$\sim 10^7$	27.72	— (deeper than every class)

D.3. Motif and Flank Perturbation Summary

Table A12 summarises the motif-edit and flank-shuffle controls of §3.2. On 1,000 canonical GT-AG donors (chr22, ± 10 bp pad-averaged c) the central GT motif contributes a small but reliable detection signal ($\bar{c}$ deepens by 0.46 layers when GT is replaced with AA), whereas dinucleotide-shuffling the ± 100 bp flank lifts $\bar{c}$ to a shallower value by 3.18 layers, isolating the GT for easier stabilisation. Real flanking grammar therefore demands deeper integration than the central motif on its own.

Table A12. Motif-edit and flank-shuffle controls on 1,000 canonical GT-AG donors. Real $\bar{c} = 26.77$. Replacing the central GT with AA (while keeping the flank) deepens $\bar{c}$; shuffling the flank (while keeping GT) lifts $\bar{c}$ to a shallower value by 3.18 layers.

Condition	$\bar{c}$	$\Delta \bar{c}$ vs. real	Cohen’s d	paired Wilcoxon p
real GT-AG donor (reference)	26.77	0 (ref.)	—	—
GT $\rightarrow$ AA, flank preserved	27.24	+0.46	−0.086	2.3×10^{-32}
GT preserved, ± 100 bp flank shuffled	23.59	−3.18	+0.515	4.1×10^{-59}

E. Entropy Decoupling at Genome-Wide Scale

The body of the paper treats $c(t)$ as decoupled from per-position next-token entropy H_t at all canonical contexts except 5’ UTR (§3.1). Here we extend the diagnostic from the 120-window control panel of §3.1 to the full chr22 analysis windows (12,978 windows, 38.9 M analysed positions) to confirm that the decoupling argument is not an artefact of the small control sample. We use the canonical $c(t)$ from the chr22 forward joined to a per-position H_t computed from the post-norm logits on the same chr22 sequence. Table A13 reports the per-context joint $(n, \bar{c}, \sigma(H_t), \text{IQR}(H_t), \rho(c, H_t))$.

Two points are worth noting. First, $|\rho(c, H_t)| \leq 0.224$ across all seven canonical contexts at this scale; the entropy decoupling argument therefore holds genome-wide. The largest residual couplings appear at splice acceptor ($\rho = -0.224$) and intron ($\rho = -0.149$); these are negative, indicating that within these contexts higher prediction confidence is associated with shallower settling. This direction is consistent with the splice signal being driven by representational stability rather than by confidence inflation. Second, 5’ UTR yields $\rho = +0.054$ at $n = 74,728$ positions, smaller than the +0.41 observed on the 120-window control sample. We interpret the control-panel value as a small-sample property of the specific window draw rather than a stable biological feature, and continue to keep 5’ UTR on a separate axis in Fig. 2 as the most conservative reporting choice. A simple $\sigma(H_t)$ -driven restriction-of-range explanation (whereby narrow entropy spread mechanically suppresses ρ) is not supported by the data: 5’ UTR has the narrowest $\sigma(H_t)$ in the table yet does not produce the smallest $|\rho|$.

Table A13. Per-context joint distribution of canonical settling depth $c(t)$ and per-position next-token entropy H_t on the full chr22 analysis windows (38.9 M positions; central 3 kb of each 12,978 window). $\sigma(H_t)$ and $\text{IQR}(H_t)$ measure per-context entropy spread; $\rho(c, H_t)$ is Spearman correlation. Contexts are ordered by $|\rho|$.

Context	n	$\bar{c}$	$\sigma(H_t)$	$\text{IQR}(H_t)$	$\rho(c, H_t)$
splice acceptor	93,883	25.69	0.496	0.944	−0.224
intron	20,595,963	27.82	0.395	0.563	−0.149
splice donor	94,528	25.59	0.507	0.982	−0.134
coding exon	1,643,533	28.25	0.504	1.009	−0.097
5’ UTR	74,728	29.03	0.366	0.336	+0.054
3’ UTR	610,330	27.73	0.357	0.226	−0.052
intergenic	15,821,035	28.75	0.468	0.918	+0.001

F. Single-Base Promoter Elements

To test whether the splice-site shallow-settling signature of §3.1 generalises to other single-base regulatory anchors, we measure canonical $c(t)$ at four promoter-element classes on chr22: canonical transcription start sites (TSS, GENCODE v44 Ensembl-canonical), and JASPAR position-weight-matrix hits for TATA-box (MA0108.3, TBP), CAAT-box (MA0060, NFY), and GC-box (MA0079, Sp1). PWM hits are restricted to positions within ± 1.5 kb of a canonical TSS so that each anchor lies in a functional promoter context. Because promoter elements are composition-loaded (GC-box is GC-rich by definition), we report Cohen’s d against two baselines: the chr22 intron baseline used elsewhere in the paper, and a GC-content-matched non-promoter chr22 control (5%-wide GC bin, 1:1 matched). Table A14 reports the resulting effect

sizes.

The splice-shallow signature does not generalise to single-base promoter elements. None of TSS, TATA, CAAT, or GC-box settles shallower than the intronic baseline; TATA and GC-box settle measurably deeper ($d = +0.22$, $+0.29$ vs. intron). Under the GC-content-matched control, only CAAT-box exhibits a small shallower-settling shift ($d = -0.12$), and GC-box remains robustly deep ($d = +0.31$). TATA and TSS are statistically indistinguishable from their composition-matched controls ($|d| \leq 0.05$), so the apparent depth shift versus intron in those rows is largely a composition effect rather than a residual-stream signature of motif detection. We read this as direct support for arm (b) of the bidirectional metric: at promoter-element grammars, where Evo 2 must integrate CpG-island and TF-binding-site context, $c(t)$ shifts deeper rather than shallower. The splice-shallow signal of §3.1 therefore reflects a splice-specific motif-detection arm (arm (a)) rather than a generic short-motif phenomenon.

Table A14. Canonical settling depth at chr22 promoter elements. “vs. intron” is Cohen’s d against the chr22 intron baseline; “vs. GC-matched control” is Cohen’s d against a chr22 non-promoter control sampled in the same $\pm 5\%$ GC bin (1:1 matched). For reference, the chr22 splice-donor and splice-acceptor anchors yield $d = -0.367$ and -0.351 on the same intron baseline.

Promoter element	n	$\bar{c}$	d vs. intron	d vs. GC-matched ctrl
TSS (Ensembl-canonical)	1,410	28.33	+0.085	+0.042
TATA-box (MA0108.3, TBP)	13,560	29.14	+0.219	+0.027
CAAT-box (MA0060, NFY)	4,392	28.11	+0.049	-0.121
GC-box (MA0079, Sp1)	3,901	29.56	+0.287	+0.313

F.1. Per-Context Aggregation Sanity Check

Context-level summaries in the body use per-position $c(t)$ pooled over windows: $\bar{c}_{\text{ctx}} = |S_{\text{ctx}}|^{-1} \sum_{t \in S_{\text{ctx}}} c(t)$. As a sanity check, we recomputed the context-level ranking using per-region mean c (one value per annotated region rather than per position). The Spearman correlation between the position-pooled and region-pooled per-context $\bar{c}$ vectors is ≥ 0.97 on both chr22 and chr17, and the rank ordering of the seven contexts is preserved. We report d throughout because it normalises by pooled within-context standard deviation and so absorbs within-region variability by construction; both aggregation schemes lead to the same qualitative ordering.

G. Reproducibility

All experiments use random seed 42 for cross-validation splits and bootstrap resamples. The Evo 2 model lock is `arcinstitute/evo2.7b` at HF revision SHA `bda0089f92582d5baabf0f22d9fc85f3588f6b58` (weights MD5 `359ef88ccac2a62644035578de8a7db4`). Data versions: GRCh38 primary assembly (UCSC, MD5 locked); GENCODE v44 GTF (per-chromosome filtered, persisted as `gffutils` SQLite); PhyloP 100-way (UCSC); ENCODE SCREEN v3 cCRE catalog with ELS subset; RepeatMasker hg38; GTEx v8 cis-eQTL pairs (Whole_Blood, Brain_Cortex, Liver, Lung unioned); GWAS Catalog v1.0; ClinVar 2026-04-18. Software stack: `torch 2.4.1+cu124`, `evo2 0.3.0`, `vortex 1.0.8`, `transformer-engine 2.14.0`, `transformers 4.49.0`, `scipy 1.13`, `scikit-learn 1.4`. Hardware: a single NVIDIA H200 (141 GB). Total compute: ~ 20 GPU-hours end-to-end. Code, dataset version locks, and figure-generation scripts will be released at an anonymous URL upon acceptance (an anonymized repository is provided as supplementary material for review).